DANIEL KARIMI

TELECOMMUNICATIONS ENGINEER & SOFTWARE DEVELOPER

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ABOUT ME

Telecommunications Systems Engineer graduated from **Universidad Politécnica de Madrid**, with a strong background in communication systems, networking, and software development. My approach combines technical skills in **programming**, **data processing**, and **cloud & DevOps tools** with a problem-solving mindset oriented towards building reliable, scalable solutions.

I am passionate about **low-level Artificial Intelligence**: from training and developing my own LLM models to implementing software solutions that solve both technical and everyday problems. I firmly believe in understanding how things work from their foundations.

I am looking for opportunities where I can combine my knowledge in **telecommunications** with the development of **innovative AI and software solutions**, creating technology that has a real impact on the world.

TECHNICAL STACK

💻 PROGRAMMING

Java • Python • JavaScript / TypeScript • C • SQL • Scala (Basic)

☁️ CLOUD & DEVOPS

Docker • Kubernetes • CI/CD Pipelines • GitHub Actions • Vercel • Cloudflare Workers • Linux • Git

🗄️ DATA & PLATFORMS

Data Pipelines • Data Curation • REST APIs • PostgreSQL • Pandas • NumPy • Jupyter

🌐 WEB DEVELOPMENT

React.js • Next.js • Node.js • HTML5 / CSS3

🤖 AI & COMPUTER VISION

YOLO (v5/v8) • PyTorch • OpenCV • Deep Learning • TensorFlow • Image Processing

🛠️ TOOLS & METHODS

Git / GitHub • VS Code • Agile Methodologies • Technical Documentation • Application Monitoring

Languages: Spanish (Native) • English (B2 – Aptis ESOL, British Council)

PROFESSIONAL EXPERIENCE

AI Engineer Intern | CSIC – Institute of Physical and Information Technologies (ITEFI)

Oct 2025 – Jan 2026

- Developed automated **data processing pipelines** for biological image analysis: conversion, normalization, and optimization of large datasets.
- Built and maintained Computer Vision systems using **YOLOv8** and **PyTorch**, including dataset generation, curation, and quality assurance.
- Configured and managed development environments (**Anaconda/Python**) and resolved dependency issues across multiple platforms.
- Collaborated with the RESULT research group, translating complex requirements into data structures and engineering solutions.
- Performed **performance analysis and optimization** of ML models: metrics evaluation and improvement of predictive accuracy.
- Created comprehensive **technical documentation** for tools, processes, and workflows.

Python | Data Pipelines | PyTorch | OpenCV | Documentation | Problem Solving

Telecommunications Systems Engineering | Universidad Politécnica de Madrid (ETSIT-UPM)

Sep 2020 – Jul 2025

- **Programming & Algorithms:** Data structures, OOP, algorithm design and optimization (C, Python, Java).
- **Networking & Systems:** TCP/IP architecture, network protocols, operating systems, distributed systems.
- **Digital Signal Processing:** Spectral analysis, digital filtering, real-time DSP algorithms.
- **Electronics & Embedded Systems:** Microprocessors, analog/digital circuits, hardware-software integration.

Java | C | Python | Networking | DSP | Algorithms

EDUCATION

Bachelor's Degree in Telecommunications Systems Engineering, Universidad Politécnica de Madrid (ETSIT-UPM) *2020 – 2025*

Comprehensive training in communication systems, digital signal processing, electronics, networking, and software development. Specialization in AI and Computer Vision.

Technology Baccalaureate, IES Isabel la Católica *2018 – 2020*

Science and Technology track. Honours. Strong foundation in mathematics and physics.

CERTIFICATIONS

AI and Productivity (Google): Efficient use of AI tools and automation workflows. •

Generative AI (MIT): Implementation of GenAI solutions and sentiment analysis. •

Internet of Things (MIT): IoT solutions for smart homes and precision agriculture. •

Digital Transformation (MIT): Technology strategy and business digitalization. •

Aptis ESOL B2 (British Council): Official English certification accrediting full linguistic competence (CEFR). Assessment of Reading, Writing, Listening and Speaking, demonstrating ability to work in professional environments and understand complex technical language.

KEY PROJECTS

CellTracker AI + SCIENTIFIC PAPER

PYTHON • YOLOV8 • PYTORCH • OPENCV

AI-powered cell tracking system for microscopy. Pending publication in scientific journal.

Computer Vision project for biology labs. Implements custom YOLOv8-based detection models for automatic cell identification and tracking. Includes a complete image processing pipeline, multi-object tracking with Kalman filters, and statistical metrics export.

- Custom-trained YOLOv8 detection and multi-object tracking (Kalman Filters).
- End-to-end data pipeline: preprocessing, quantitative trajectory analysis, and metrics export.

MigraCell Analyzer DEEP LEARNING + BIO

PYTHON • YOLOV8-SEG • PYTORCH • TKINTER

Intelligent Microscopy Platform for Cell Migration Analysis.

Full automation of wound healing assays using Deep Learning. Processes complete experiments in < 5 minutes (vs 4-8h manually). Inference with YOLOv8-seg, mask post-processing, geometric computation, and automatic report generation.

- 100% reproducible results with pixel-perfect YOLOv8-seg segmentation.
- Unattended batch processing and publication-ready chart generation.